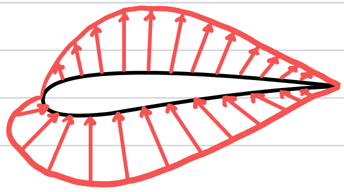
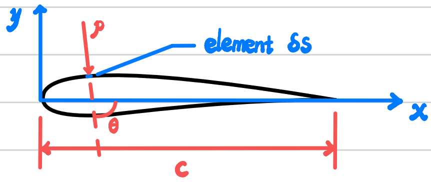
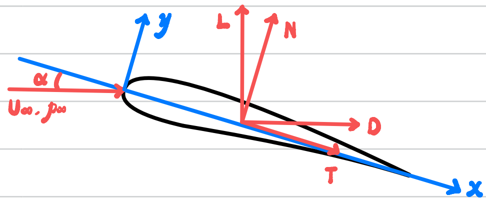
12/10/2023 Profile Drag and Wake Momentum Pre-lab

Theory

Pressure (form) drag pressure distribution around airfoil







2-D

$T = p\delta s \cos\theta$ and $N = p\delta s \sin\theta$

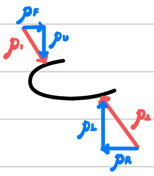
2-D

$D = T\cos\alpha + N\sin\alpha$

$T = \oint p\cos\theta\, ds$ and $N = \oint p\sin\theta\, ds$

$\cos\theta\, ds = dx$

$\sin\theta\, ds = dy$



$C_p = \dfrac{p - p_\infty}{\frac{1}{2}\rho U_\infty^2}$

$C_D = \dfrac{D}{\frac{1}{2}\rho U_\infty^2 c}$

$T = \int (p_F - p_R)\, dx$ and $N = \int (p_L - p_U)\, dy$

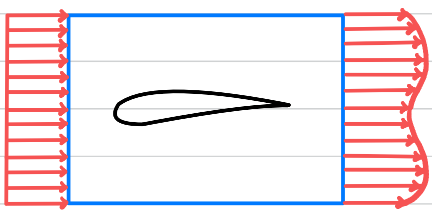
symmetric wing with $\alpha = 0$

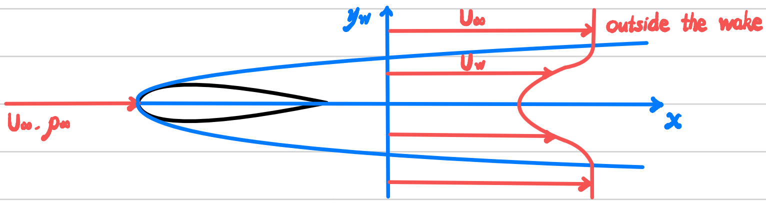
$p_U = p_L$ and $N = 0$ and $T = D$

$C_D = 2\int_0^Y (C_{p,F} - C_{p,R})\, d(\frac{y}{c})$ $\quad * Y = \dfrac{\text{maximum thickness}}{c}$

Skin friction drag

Profile drag $D_{profile} = D_p + D_{skin\ friction}$ momentum deficit in the wake





for steady flow $\xrightarrow{\text{C.V.}}$ $0 +$ net momentum out of C.V. $= 0 + \rho\int U_w (U_w - U_\infty)\, dy_w = -D$

$\xrightarrow{C_D = \frac{D}{\frac{1}{2}\rho U_\infty^2 c}}$ $C_D = \dfrac{-\rho\int U_w (U_w - U_\infty)\, dy_w}{\frac{1}{2}\rho U_\infty^2 c} = \dfrac{2}{c}\int \dfrac{U_w}{U_\infty}\left(1 - \dfrac{U_w}{U_\infty}\right) dy_w$

$\xrightarrow[p_0 = p + \frac{1}{2}U^2 \text{ where } p = p_\infty]{\text{total pressures } H_w \text{ and } H_\infty}$ $C_D = \dfrac{2}{c}\int \left(\dfrac{H_w - p_\infty}{H_\infty - p_\infty}\right)^{\frac{1}{2}} \left[1 - \left(\dfrac{H_w - p_\infty}{H_\infty - p_\infty}\right)^{\frac{1}{2}}\right] dy_w$

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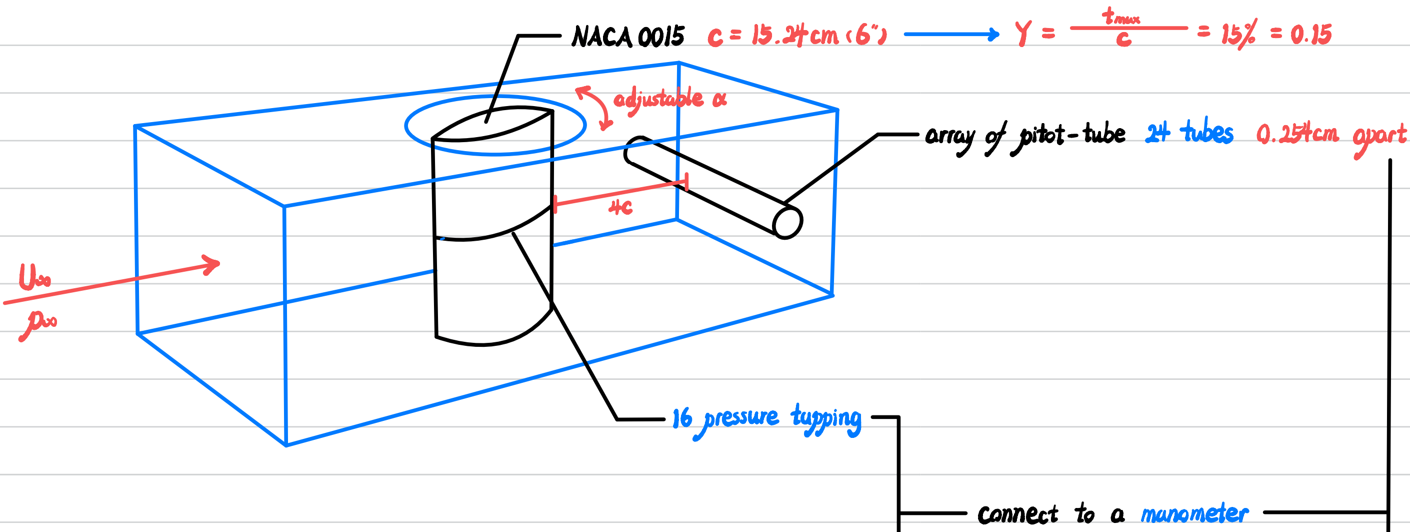
Procedure

Appratus



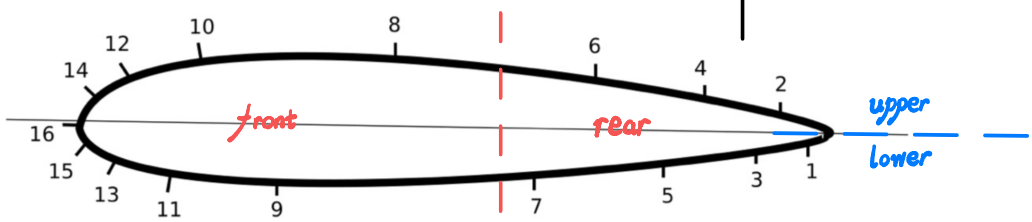


Figure 3: Location of pressure tappings

Hole No.	**1**	**3**	**5**	**7**	**9**	**11**	**13**	**15**
x/c	0.958	0.858	0.713	0.525	0.152	0.065	0.020	0.003
y/c	0.0094	0.02815	0.0469	0.06565	0.06565	0.0469	0.02815	0.0094
Hole No.	**16**	**14**	**12**	**10**	**8**	**6**	**4**	**2**
x/c	0	0.0087	0.028	0.090	0.303	0.593	0.742	0.888
y/c	0	0.01875	0.0375	0.05625	0.075	0.05625	0.0375	0.01875

Table 1: Surface Pressure Tappings

Tube No.	**1**	**3**	**5**	**7**	**8**	**9**	**10**	**11**	**12**
Distance from Tube 1 [cm]	0	0.508	1.016	1.524	1.778	2.032	2.286	2.54	2.794
Tube No.	**13**	**14**	**15**	**16**	**17**	**18**	**20**	**22**	**24**
Distance from Tube 1 [cm]	3.048	3.302	3.556	3.81	4.064	4.318	4.826	5.334	5.842

Table 2: Wake Pitot Tube Array

Procedure

① $\alpha = 0°$, $\theta_{manometer} = 0°$ $\xrightarrow{U_\infty \uparrow}$ $h_{manometer} = 55$ mm WC

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② record data

③ plot h to see if it is symmetric

Error Analysis

Theory

Assume independent variable yields a common formula

$$s_f = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 s_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 s_y^2 + \left(\frac{\partial f}{\partial z}\right)^2 s_z^2 + \dots}$$ where s represent standard deviation

for absolute error

$$\Delta e_f = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 \Delta e_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \Delta e_y^2 + \left(\frac{\partial f}{\partial z}\right)^2 \Delta e_z^2 + \dots}$$

Code provided by

```
function summation = ErrorPropagation(f,variables,values,errs)









end
```

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Procedure

Checklist

① pressure tapping align with wake pitot tube array

② $\alpha = 0^\circ$, $\theta_{manometer} = 0^\circ$ $\xrightarrow{U_\infty \uparrow}$ $h_{manometer} = 55\,mm\ WC$

③ change manometer angle each time

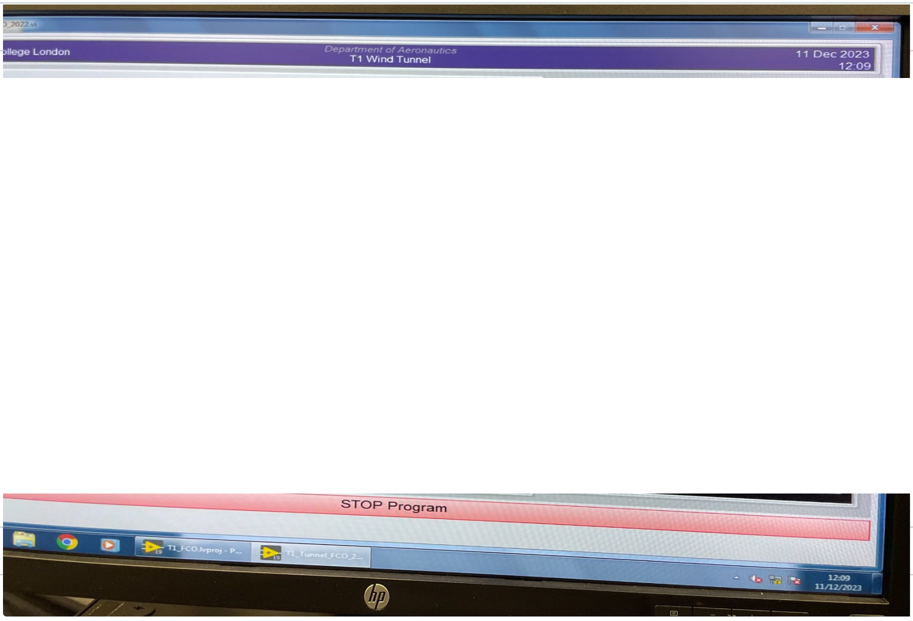
④ plot manometer reading to make sure it is symmetric ⟶ change if it is not

⑤ find source of error

Data recording

Computer reading

Machine	U_∞ (m/s)	$\rho_{alcohol}$ (kg/m^3)	T (°C)	T (K)	$P_{atmosphere}$ (kPa)	$h_{dynamic}$ (mm H2O)



Machine	$\theta_{reading}$ (±°)	$\Delta\theta$ (±rad)	$T_{reading}$ (±°C)	$T_{experiment}$ (±°C)	ΔT_{total} (±°C)	$\Delta h_{dynamic}$ (±mm H2O)	$\Delta h_{dynamic}$ (±m)

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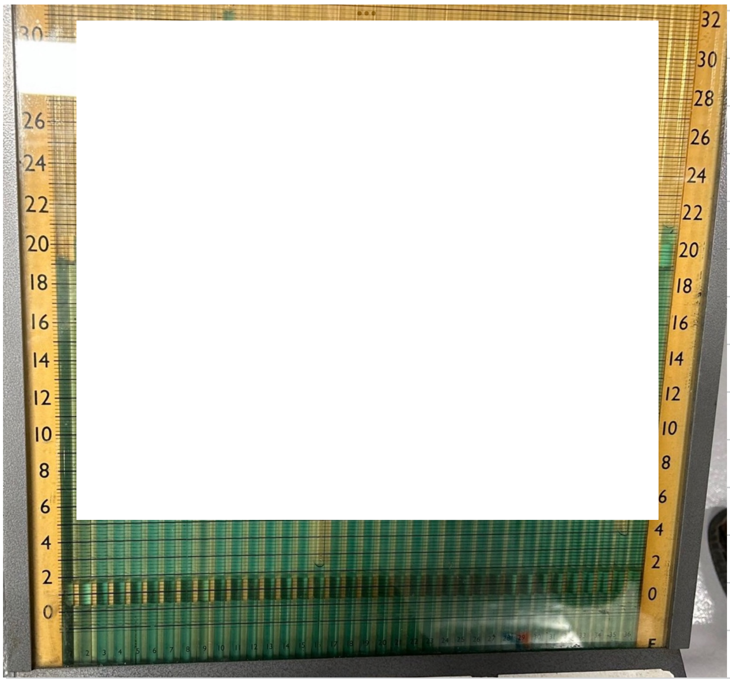
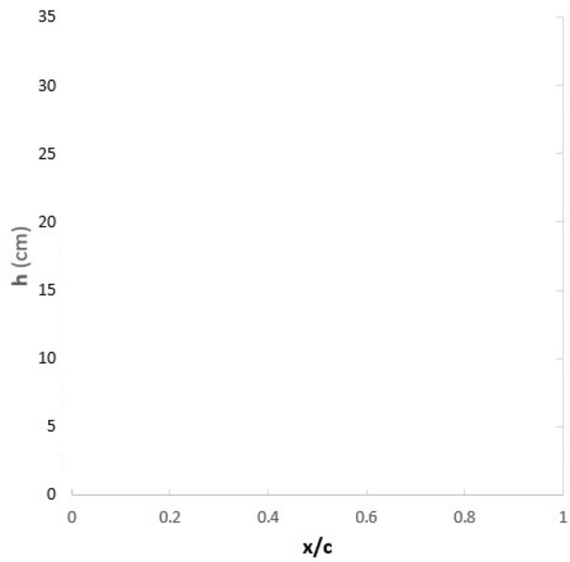
Wing

Manometer		θ (°)	θ (rad)	h (cm)	h (m)	x/c	y/c
Surface Pressure Tapping							





Manometer		$h_{reading}$ (±cm)	$h_{experiment}$ (±cm)	Δh_{total} (±cm)	Δh (±m)
Surface Pressure Tapping	1				
	2				
	3				
	4				
	5				
	6				
	7				
	8				
	9				
	10				
	11				
	12				
	13				
	14				
	15				
	16				

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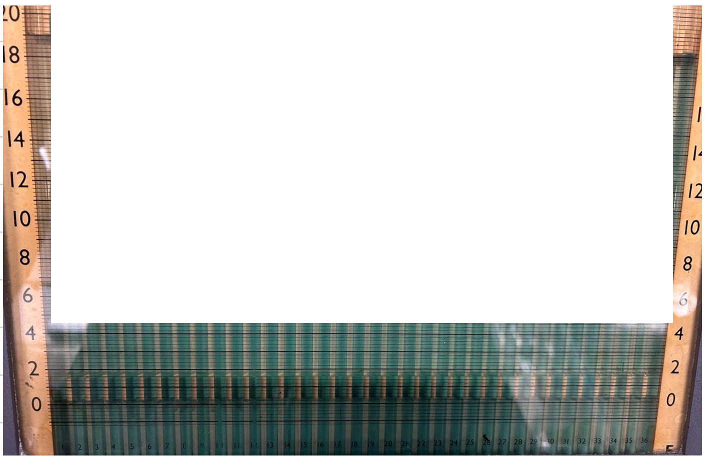
Wake

Manometer		θ (°)	θ (rad)	h (cm)	h (m)	x/c	y/c
Surface Pressure Tapping	1						
	2						
	3						
	4						
	5						
	6						
	7						
	8						
	9						
	10						
	11						
	12						
	13						
	14						
	15						
	16						
	Static						
	Total						



Manometer		$h_{reading}$ (±cm)	$h_{experiment}$ (±cm)	Δh_{total} (±cm)	Δh (±m)
Wake Pitot Tube Array	1		[illegible]	[illegible]	[illegible]
	3				
	5				
	7				
	8				
	9				
	10				
	11				
	12				
	13				
	14				
	15				
	16				
	17				
	18				
	20				
	22				
	24				

12/12/2023 # Profile Drag and Wake Momentum Post-lab

Theory

Pressure drag

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho U_\infty^2} = \frac{\rho_{alcohol}\, g\,[-(h - h_{static})]}{\rho_{water}\, g\, h_{dynamic}} = \frac{\rho_{alcohol}\, g \sin(\theta)(h_{static} - h_{aerofoil})}{\rho_{water}\, g\, h_{dynamic}}$$

Profile drag

$$\left(\frac{H_W - p_\infty}{H_\infty - p_\infty}\right)^{\frac{1}{2}}\left[1 - \left(\frac{H_W - p_\infty}{H_\infty - p_\infty}\right)^{\frac{1}{2}}\right] = \left(\frac{\rho_{alcohol}\, g \sin(\theta)(h_{static} - h_{wake})}{\rho_{water}\, g\, h_{dynamic}}\right)^{\frac{1}{2}}\left[1 - \left(\frac{\rho_{alcohol}\, g \sin(\theta)(h_{static} - h_{wake})}{\rho_{water}\, g\, h_{dynamic}}\right)^{\frac{1}{2}}\right]$$

$$\therefore\ C_D = \frac{2}{c}\sum\left(\frac{H_W - p_\infty}{H_\infty - p_\infty}\right)^{\frac{1}{2}}\left[1 - \left(\frac{H_W - p_\infty}{H_\infty - p_\infty}\right)^{\frac{1}{2}}\right] d$$

Skin friction drag

$$C_{Df} = \sqrt{C_D^2 - C_{Dp}^2}$$

Data and Error Analysis

Pressure drag and profile drag integral calculation

		Recording				Calculation				
Machine		U_∞ (m/s)	T (K)	$P_{atmosphere}$ (Pa)	$h_{dynamic}$ (m)	$\Delta P_{dynamic}$ (Pa)	ρ_{air} (kg/m^3)			
Manometer		θ (rad)	h (m)	x/c	y/c	$\Delta P_{airfoil}$ (Pa)	ΔP_{static} (Pa)	C_p		
Surface Pressure Tapping	1									
	2									
	3									
	4									
	5									
	6									
	7									
	8									
	9									
	10									
	11									
	12									
	13									
	14									
	15									
	16									
	Static									
Manometer		θ (rad)	h (m)	d (m)		ΔP_{wake} (Pa)	ΔP_{static} (Pa)	$H_w - P_\infty$ (Pa)	$H_w - P_\infty / H_\infty - P_\infty$ (Pa)	Integrand
Wake Pitot Tube Array	1	[illegible]	[illegible]	[illegible]		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
	3									
	5									
	7									
	8									
	9									
	10									
	11									
	12									
	13									
	14									
	15									
	16									
	17									
	18									
	20									
	22									
	24									
	Total									

12/12/2023 Profile Drag and Wake Momentum Post-lab

C_p vs $\frac{x}{c}$

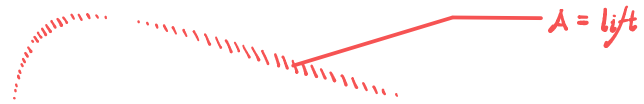


Figure 6. -Cp v.s. x/c

C_p vs $\frac{y}{c}$

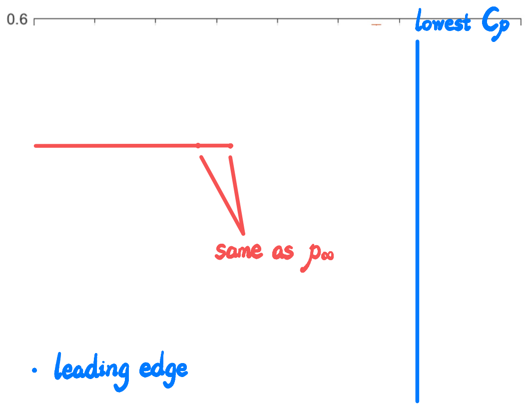
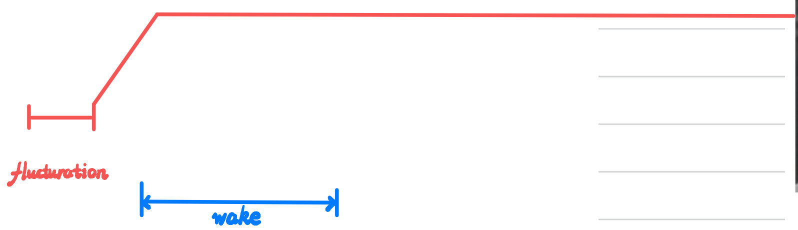


Figure 7. -Cp v.s. y/c

Integrand vs y_w



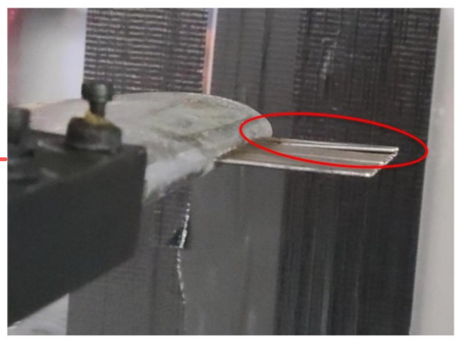


Figure 9. Bent pitot tube in wake array

Figure 8. $\left(\frac{H_W-p_\infty}{H_\infty-p_\infty}\right)^{\frac{1}{2}}\left(1-\left(\frac{H_W-p_\infty}{H_\infty-p_\infty}\right)^{\frac{1}{2}}\right)$ vs y_w

12/12/2023 **Profile Drag and Wake Momentum** Post-lab

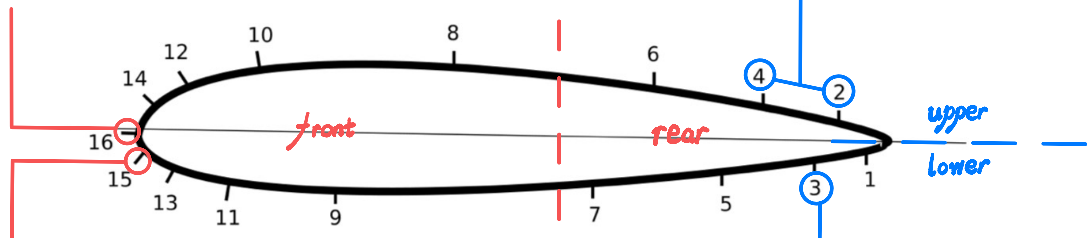
Results

		$C_{D\ Profile}$	$C_{D\ Pressure}$	$C_{D\ Skin\ Friciton}$	C_L
Experiment	result				
	error				
CFD					

CAD model not smooth

max UC_p = ±0.0454 (min %UC_p = 4.38%)

min UC_p ±0.008 (max %UC_p = 13.56%)



min UC_p ±0.0077 (max %UC_p = 35.62%)

Figure 3: Location of pressure tappings

max UC_p = ±0.0077 (max %UC_p = 4.43%)

Hole No.	1	3	5	7	9	11	13	15
x/c	0.958	0.858	0.713	0.525	0.152	0.065	0.020	0.003
y/c	0.0094	0.02815	0.0469	0.06565	0.06565	0.0469	0.02815	0.0094
Hole No.	**16**	**14**	**12**	**10**	**8**	**6**	**4**	**2**
x/c	0	0.0087	0.028	0.090	0.303	0.593	0.742	0.888
y/c	0	0.01875	0.0375	0.05625	0.075	0.05625	0.0375	0.01875

Table 1: Surface Pressure Tappings

min UC_{po} = ±0.0120 (min %UC_{po} = 17.31%)

Tube No.	1	3	5	7	8	9	10	11	12
Distance from Tube 1 [cm]	0	0.508	1.016	1.524	1.778	2.032	2.286	2.54	2.794
Tube No.	**13**	**14**	**15**	**16**	**17**	**18**	**20**	**22**	**24**
Distance from Tube 1 [cm]	3.048	3.302	3.556	3.81	4.064	4.318	4.826	5.334	5.842

max UC_{po} = ±0.0144 (max %UC_{po} = 95.45%)

Table 2: Wake Pitot Tube Array

Blockage effects

blockage ratio $= \dfrac{A_{airfoil}}{A_{wind\text{-}tunnel}} \times 100\% = \dfrac{0.0113\,m^2}{0.451\,m^2} \times 100\% = 2.51\% < 5\% \longrightarrow$ insignificant

Possible Error Sources

- non-uniform surface roughness $\longrightarrow$ L $\longrightarrow$ lift induced drag $\longrightarrow C_{D,p} \uparrow$
- change T $\longrightarrow$ change μ and ρ_{air}
- pitot-tube not aligned
- ground not leveled
- parallex error when read manometer